

Secret Communication System and Cola Tablets

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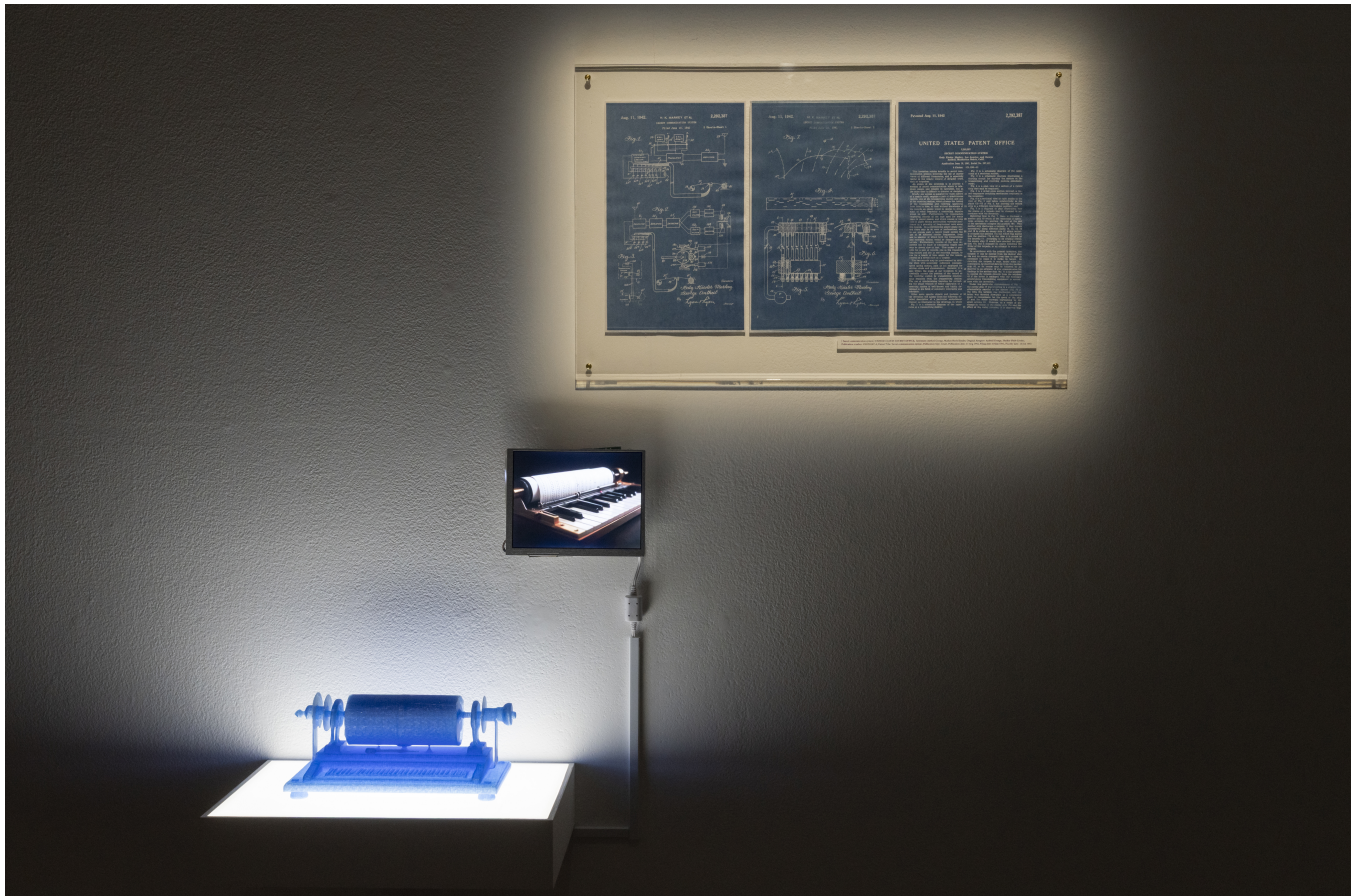


Figure 1: The installation of *Secret Communication System and Cola Tablets* at Shiseido Gallery 2025. Photo by Ken Kato

Abstract

This artwork reimagines the unrealized inventions of Hollywood actress and inventor Hedy Lamarr, using generative AI and 3D printing to speculate on women's contributions to science and technology that have been historically overlooked due to gender bias.

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Lamarr, an inventor in addition to being a famous Hollywood actress, co-developed and patented a “Secret Communication System” in 1942 with composer George Antheil. This invention, based on the mechanism of a player piano, was a technique for secure radio control of torpedoes. Known as “frequency-hopping spread spectrum,” it laid the groundwork for modern technologies like Wi-Fi, Bluetooth, and GPS.

However, the Navy rejected their invention at that time, reportedly because they were unwilling to accept an invention by a woman. Also, when she wanted to register with the National Inventors Council, she was told to use her beauty to sell war bonds instead.

Her communication technology was eventually put into military use in the 1960s, but her contribution went unrecognized for

decades. It was not until 1994, with the standardisation of Wi-Fi, that she finally received recognition, underscoring how her gender made her contributions invisible.

In interviews, Lamarr also spoke of other inventions, including a cube-shaped cola tablet. Since these inventions didn't come to fruition, this project aimed using generative AI to recreate her invention, referring to the patent documents, photos and ads she promoted to create speculative images. These images are then used to generate 3D models and printed as physical objects.

The project uses AI as a tool for speculative history, challenging the gap of "realized" and "unrealized" inventions. The goal isn't to produce accurate replicas of her patent diagrams but to use AI's interpretive and unpredictable nature to generate more poetic forms. By employing diverse sources, AI is prompted to reimagine Lamar's ideas, generating narratives over functional ones.

CCS Concepts

• **Human-centered computing** → **Gender**; • **Applied computing** → *Media arts*; • **Computing methodologies** → Artificial intelligence.

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